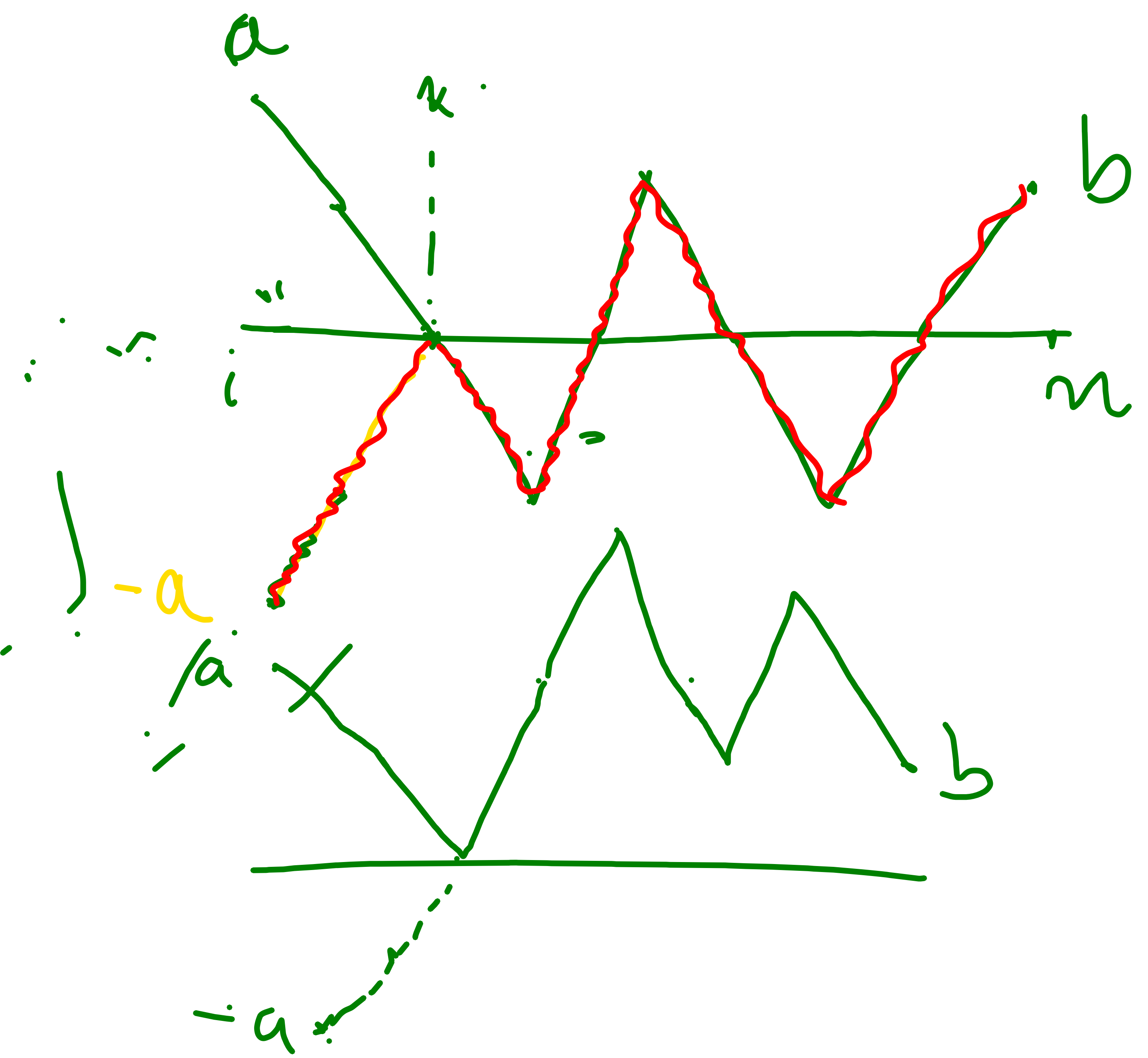
Random Walk:

<u>Thm</u>:- Prob of no return until time $2n$ is the same as that of prob of return at time $2n$.

$$P(S_1 \neq 0, \ldots, S_{2n} \neq 0) = u_{2n} = \binom{2n}{n} 2^{-2n}$$
$$= P(S_{2n} = 0)$$

Reflection Principle



$a, b > 0$

The no. of paths from (i, a) to (n, b) which touches or crosses x-axis is the same as the no. of paths from $(i, -a)$ to (n, b).

$A = \{$ paths from (i, a) to (n, b) which touches or crosses x axis $\}$.

$B = \{$ paths from $(i, -a)$ to (n, b) $\}$

$|A| = |B|$.

$\boxed{\begin{array}{l} f : A \longmapsto B. \\ f \text{ is 1-1 \& onto.} \end{array}}$

π = Path in A, $\pi = (s_i, s_{i+1}, \dots, s_n)$

$s_i = a$, $s_n = b$.

$|s_i - s_{i+1}| = 1$

Since $\pi \in A$,

$s_k = 0$ for some $i < k < n$

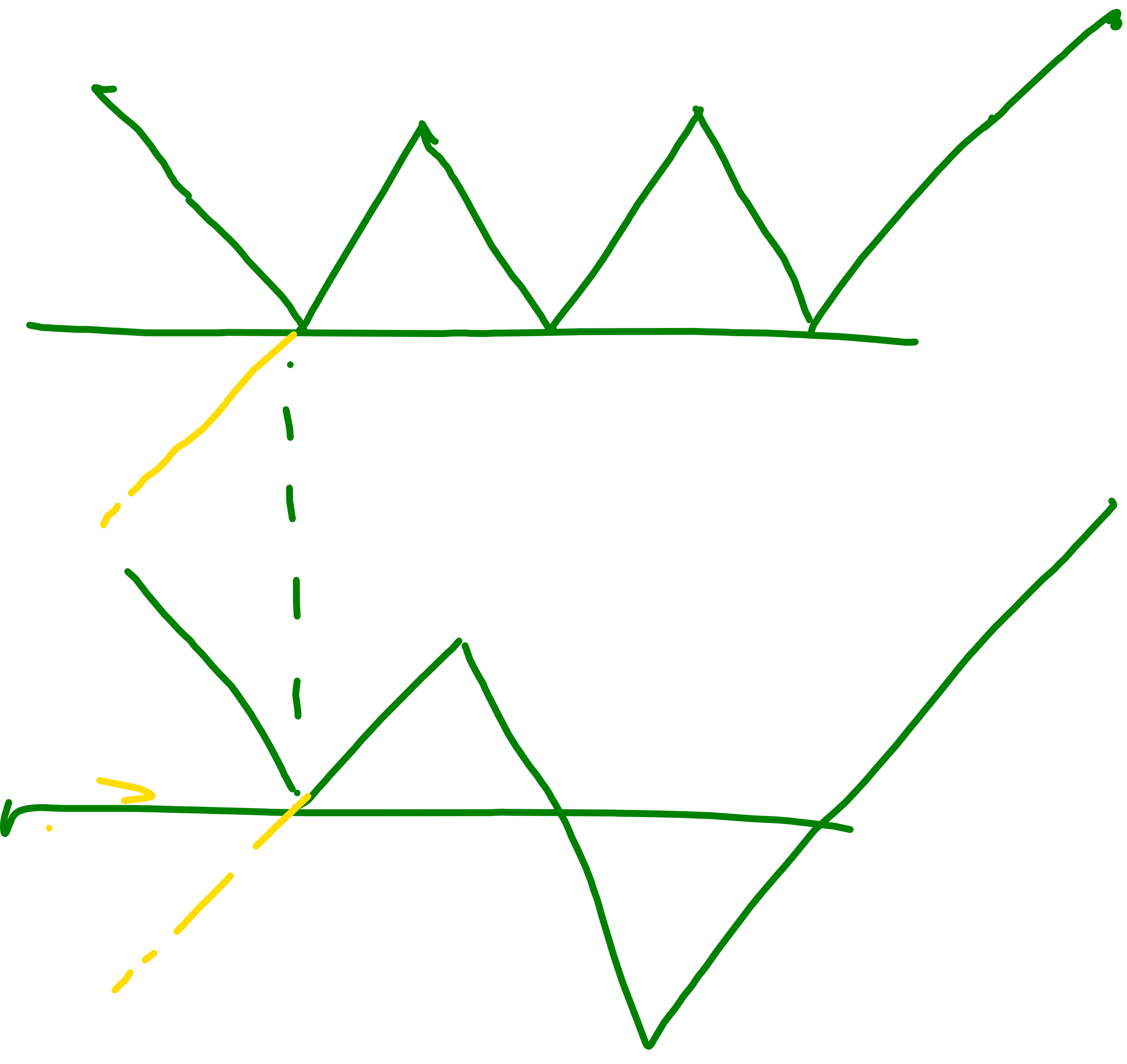
choose the smallest such k.

Now, define $\pi' = (\overbrace{-s_i, -s_{i+1}, \dots, -s_{k-1}, 0}^{\text{reflection}}, \underbrace{s_{k+1}, \dots, s_n}_{\text{original path}})$, where $0 \leftarrow s_k$.

Define $f(\pi) = \pi'$

$\pi' \in B$

Check that π' is a path starting from $-a$.



Show that f is 1-1 & onto.

Ballot Thm Two equally likely candidates A & B poll votes p & q resp. with $p>q$. Given this, the prob that candidate A has always lead over candidate B is $\dfrac{p-q}{p+q}$.

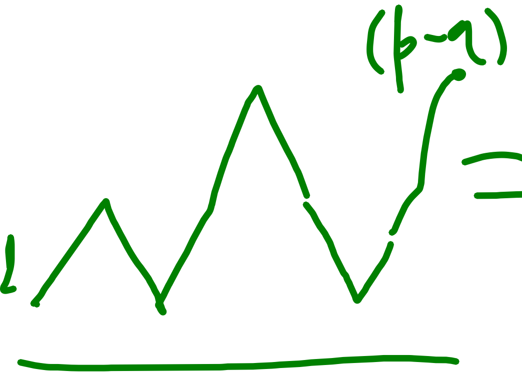
Pf:- Let $\epsilon_i = \begin{cases} +1 & \text{if } i\text{th voter votes for } A \\ -1 & \text{otherwise} \end{cases}$

$S_0 = 0$, $S_i = \sum_{1}^{i} \epsilon_k$.

S_i tracks the lead of cand A over B after i votes.

Need to show $P(S_1 > 0, S_2 > 0, \ldots S_{p+q} > 0 \mid S_{p+q} = p-q)$.

$$P(S_1>0,\dots,S_{p+q}>0 \mid S_{p+q}=p-q).$$



$$= \frac{P(S_1>0,\dots,S_{p+q}>0,\ S_{p+q}=p-q)}{P(S_{p+q}=p-q)}$$

$$= \frac{P(S_1=1,\dots,S_{p+q}=p-q)}{P(S_{p+q}=p-q)}$$

$$= \frac{\left|\{S_1=1,\dots,S_{p+q}=p-q\}\right| 2^{-(p+q)}}{P(S_{p+q}=p-q)}.$$

$$P(S_{p+q}=p-q) = \binom{p+q}{p} 2^{-(p+q)}$$

$$\left|\{S_1=1,\ldots,S_i>0,\ldots,S_{p+q}=p-q\}\right|.$$

$$= N^{(1,1)}_{(p+q,p-q)} - N^{(1,-1)}_{(p+q,p-q)}$$

Finally, $P(S_1>0,\ldots,S_{p+q}>0 \mid S_{p+q}=p-q)$.

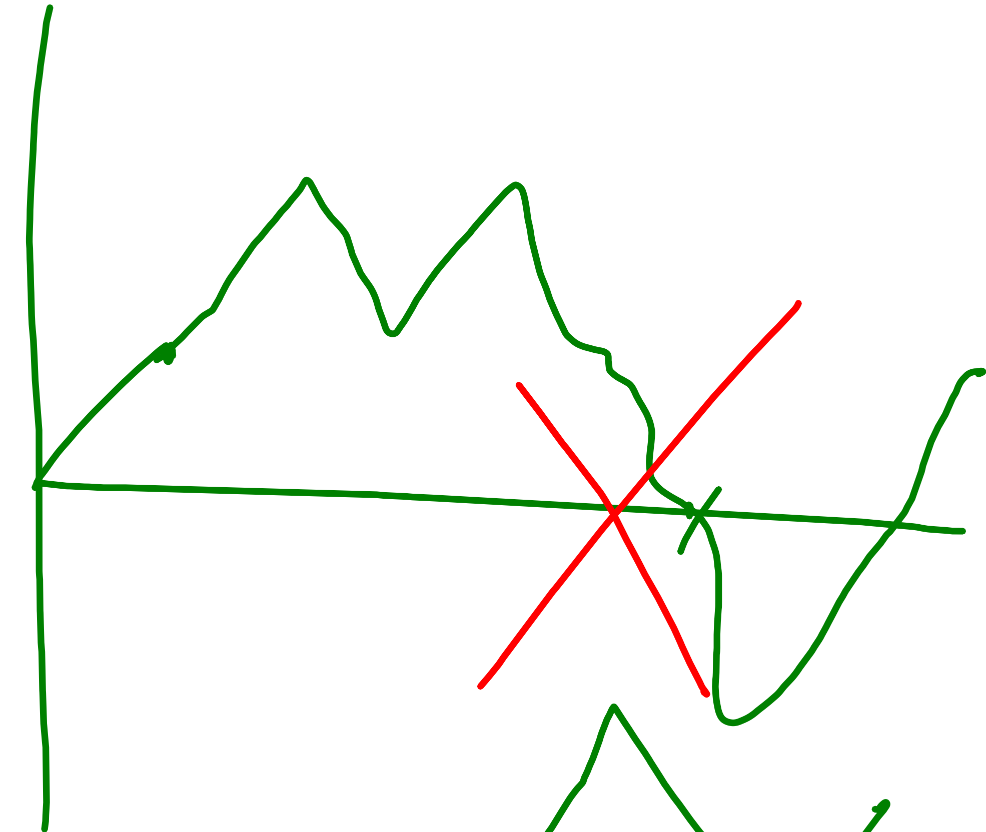
$$= \frac{N^{(1,1)}_{(p+q,p-q)} - N^{(1,-1)}_{(p+q,p-q)}}{N^{(0,0)}_{(p+q,p-q)}} = \frac{p-q}{p+q}.$$

$\boxed{\text{K}}$

$$P(S_1\neq 0,\dots,S_{2n}\neq 0)$$

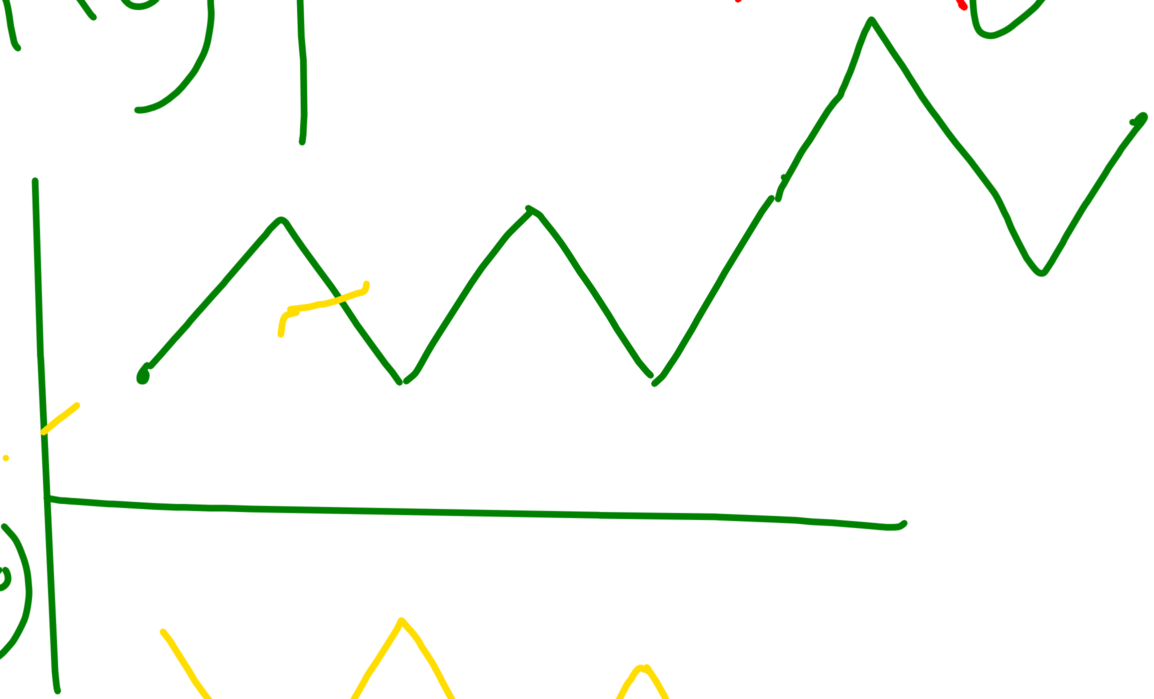
$$= P(S_1=1, S_2>0,\dots,S_{2n}>0)$$

$$+ P(S_1=-1, S_2<0,\dots,S_{2n}<0)$$



$$\epsilon_i \overset{d}{=} -\epsilon_i$$

$$= 2P(S_1=1,\dots,S_{2n}>0)$$



$$P(\epsilon_1=1, \epsilon_1+\epsilon_2>0,\dots,\epsilon_1+\dots+\epsilon_{2n}>0)$$

$$= P(-\epsilon_1=1, -\epsilon_1-\epsilon_2>0,\dots,-\epsilon_1-\epsilon_2-\dots-\epsilon_{2n}>0)$$

$$= P(\epsilon_1=-1, \epsilon_1+\epsilon_2<0, \dots)$$

$$P(\underbrace{S_1=1, S_2>0, \cdots, S_{2n}>0}_{A})$$

$$= 2^{-2n} |A|$$

$$A = \{S_1=1, S_2>0, \cdots, S_{2n}>0\}$$

$$= \bigcup_{k=1}^{\infty} \{S_1=1, S_2>0, \cdots, S_{2n}=2k\}.$$

$$= \bigcup_{k=1}^{\infty} A_k.$$

| For $k>n$, $A_k=\phi$.

$$|A_k| = N^{(1,1)}_{(2n,2k)} - N^{(1,-1)}_{(2n,2k)}$$

Convention $\binom{n}{k} = 0$ if $k > n$

$$|A| = \sum_{k=1}^{\infty} |A_k| = \sum_{k=1}^{\infty} \left[N^{(1,1)}_{(2n,2k)} - N^{(1,-1)}_{(2n,2k)} \right]$$

$$= \sum_{k=1}^{\infty} \binom{2n-1}{n+k-1} - \binom{2n-1}{n+k}$$

$$= \binom{2n-1}{n}$$

$$P(S_1=1, S_2>0, \cdots, S_{2n}>0)$$

$$= \frac{1}{2^{2n}} \cdot \binom{2n-1}{n} = \frac{1}{2^{2n}} \cdot \frac{(2n-1)!}{n!\,(n-1)!} \cdot \frac{(2n)}{n} \cdot \frac{1}{2}$$

$$= \frac{1}{2} \frac{(2n)!}{n!\,n!} \cdot 2^{-2n}.$$

$$= \frac{1}{2} u_{2n}.$$

Maxima

$$M_n = \max(\overset{=0}{S_0}, \overbrace{S_1, \dots, S_n}^{\geq 0}) \rightarrow \text{Running maxima}$$

$$P(M_n \geq k) = 2P(S_n > k) + P(S_n = k)$$

for $k = 0, 1, \dots$

Note for $k = 0$,

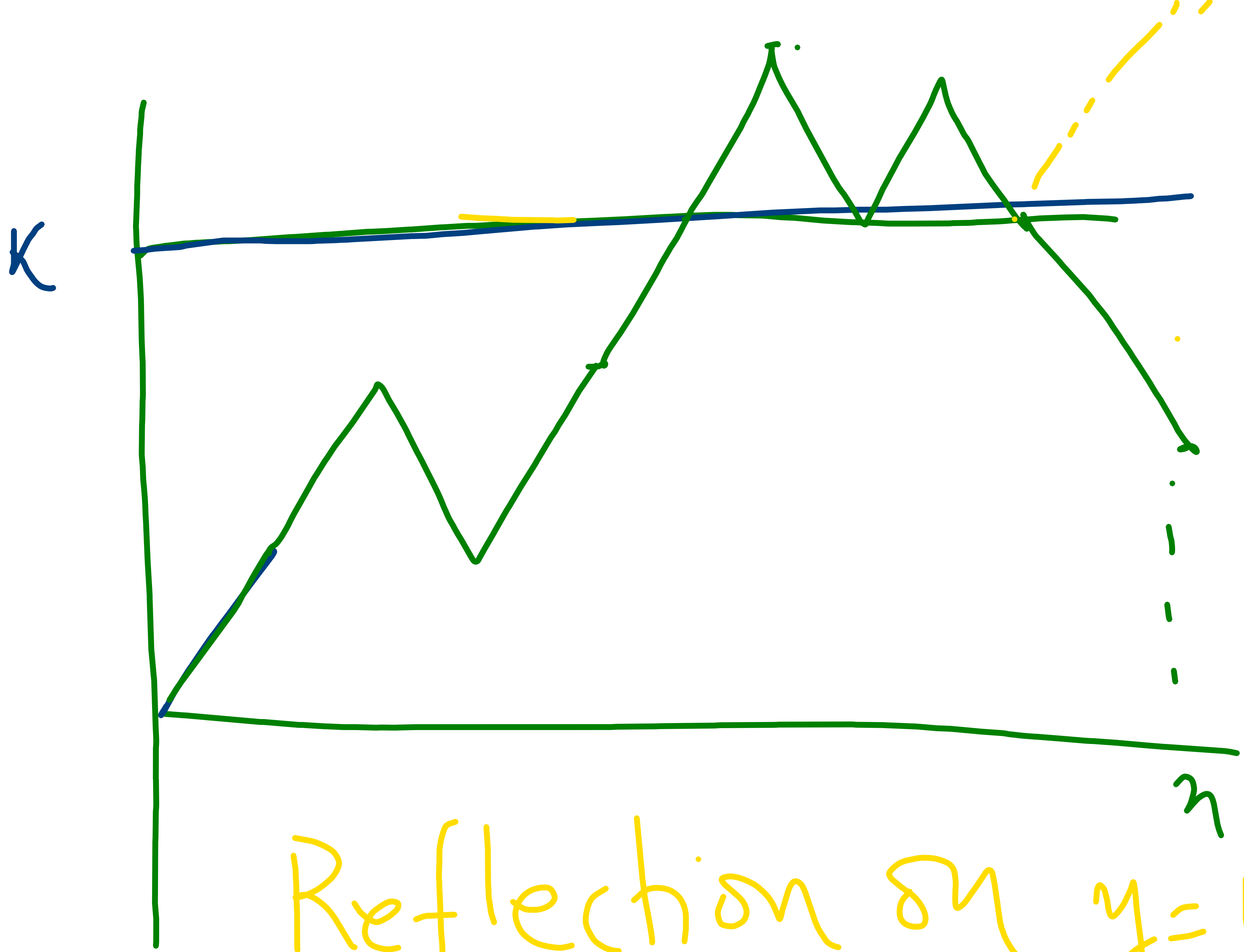
$LHS = P(M_n \geq 0) = 1$ | $RHS = 2\underbrace{P(S_n > 0)}\, + P(S_n = 0)$

$$= P(S_n > 0) + P(S_n < 0) + P(S_n = 0) = 1$$

For $k \geq 1$:

$$P(M_n \geq k) = P(\underline{M_n \geq k, S_n > k}) + P(M_n \geq k, S_n = k) + P(M_n \geq k, S_n < k)$$

$$= P(S_n > k) + P(S_n = k) + \underbrace{P(M_n \geq k, S_n < k)}_{\displaystyle \parallel \atop \displaystyle P(S_n > k)}$$



$M_0 = 0,$
$M_1 = 1,$
$M_2 = 2$
$M_3 = 2,$
$M_4 = 2$
$M_5 = 3$
$M_6 = 4$
$M_7 = \cdots 4$

Reflection on $y = k,$

$P(M_n \geq k, S_n < k)$
$= P(M_n \geq k, S_n > k) = P(S_n > k)$